

Business Requirements Document (BRD)

Title: Data Processing Steps for Customer and Order Pipelines (Delta Live Tables)

1. Executive Summary

- Purpose: Define business and functional requirements to ingest, cleanse, transform, and persist customer and order data using Delta Live Tables, produce an SCD Type 2 ordersummary, and an aggregated customeraggregatespend table.
- Outcome: Reliable Delta tables for customer, order, ordersummary (SCD2), and customeraggregatespend with defined schemas, data quality rules, transformations, and aggregations.
- Scope: Reading CSV sources, schema definitions, transformations, deduplication and null removal, SCD Type 2 implementation, aggregation, and table creation in the specified catalog/schema.

2. Business Objectives

- Enable standardized ingestion from CSV sources to Delta format.
- Maintain historical changes via SCD Type 2 for order and customer attributes in ordersummary.
- Provide aggregate spend per customer per date in customeraggregatespend.
- Ensure data quality through null removal and deduplication.

3. In-Scope

- Reading customer and order CSV data from specified volume paths into Delta tables.
- Schema definitions for customer, order, and ordersummary.
- Transformation to compute TotalAmount for orders.
- Data quality steps: null removal and deduplication for customer and order.
- Creation and maintenance of ordersummary as SCD Type 2.
- Aggregation from ordersummary to customeraggregatespend.
- Use of Delta Live Tables to orchestrate steps 1–10.

Out of Scope

- Information not present in source JSON.

4. Stakeholders and RACI

- Stakeholders: Information not present in source JSON.
- RACI:
 - Responsible: Information not present in source JSON.
 - Accountable: Information not present in source JSON.
 - Consulted: Information not present in source JSON.
 - Informed: Information not present in source JSON.

5. Current State Assessment

- Information not present in source JSON.

6. Future State Overview

- A Delta Live Tables pipeline ingests CSVs into Delta tables, applies schema enforcement, quality rules, joins customer and order to build an SCD Type 2 ordersummary with full history, and aggregates spend into customeraggregatespend for reporting.

7. Detailed Business Requirements

BR-1 Source Ingestion

- Read customer and order data from provided volume paths in CSV format and load into Delta tables.

BR-2 Schema Enforcement

- Enforce explicit schemas for customer and order as provided. Enforce defined schema for ordersummary.

BR-3 Transformation: TotalAmount

- Derive TotalAmount on order as $\text{PricePerUnit} * \text{Qty}$.

BR-4 Data Quality

- Remove Null values and nulls; deduplicate records for customer and order tables.

BR-5 Ordersummary Table Creation

- Create ordersummary table if it does not exist in catalog gen_ai_poc_databrickscoe and schema sdhc_wizard.

BR-6 Join and Load to SCD2

- Inner join customer and order on CustId; load result into ordersummary as SCD Type 2.

BR-7 SCD Type 2 Tracking

- Implement SCD2 with business keys CustId and OrderId, status column RecordStatus with values Active/Inactive, validity columns StartDate/EndDate, and rules to maintain full history.

BR-8 Aggregation to customeraggregatespend

- Aggregate sum(TotalAmount) from ordersummary grouped by Name and Date into customeraggregatespend with target columns Name, TotalAmount, Date. Create table if not exists in specified catalog/schema.

BR-9 Technology Enablement

- Implement steps 1–10 using Delta Live Tables.

8. Functional Requirements

FR-1 Source Definitions

- Sources: customer (CSV), order (CSV) at specified paths. Load targets in Delta format.

FR-2 Table Schemas

- customer: CustId, Name, EmailId, Region.
- order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.
- ordersummary: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date. (SCD2 columns RecordStatus, StartDate, EndDate also referenced in SCD2 rules.)

FR-3 Transformations

- Add column TotalAmount to order with expression PricePerUnit * Qty.

FR-4 Data Quality

- Remove values: "Null" and null. Deduplicate: true for customer and order.

FR-5 Joins

- Inner join on CustId between customer (left) and order (right).

FR-6 SCD2 Logic

- Business keys: CustId, OrderId.
- Change tracking source: customer.
- Status: RecordStatus with Active/Inactive.
- Validity: StartDate, EndDate.
- Rules per Section 9.

FR-7 Aggregations

- From ordersummary group by Name, Date; measure sum(TotalAmount); load to customeraggregatespend.

FR-8 Orchestration

- Utilize Delta Live Tables to implement steps 1–10.

9. Business Rules

- BRU-1 Null Handling: Remove rows containing "Null" (string) or null values for customer and order. Specific columns not provided in source JSON.
- BRU-2 Deduplication: Remove duplicate records from customer and order. Deduplication keys not provided in source JSON.
- BRU-3 Computation: $\text{TotalAmount} = \text{PricePerUnit} * \text{Qty}$.
- BRU-4 Join: Inner join on CustId between customer and order.
- BRU-5 SCD2 Status: When customer attributes change, mark previous ordersummary records as Inactive and new as Active; update StartDate/EndDate accordingly; maintain full history.
- BRU-6 Aggregation: For each Name and Date, $\text{TotalAmount} = \text{sum}(\text{TotalAmount})$ from ordersummary.

10. Non-Functional Requirements

- Performance: Information not present in source JSON.
- Scalability: Information not present in source JSON.
- Availability: Information not present in source JSON.
- Security and Access: Information not present in source JSON.
- Audit/Lineage: Captured via steps and Appendices; additional details not present in source JSON.
- Compliance: Information not present in source JSON.

11. Assumptions and Constraints

- Assumptions: None beyond explicit JSON content.
- Constraints: Use Delta Live Tables for implementation; target catalog/schema as specified; CSV as source format; Delta as target format.

12. Dependencies and Data Lineage

- Dependencies derived from step order and joins:
- Step 1: Sources customer and order required before downstream steps.
- Step 3 depends on Step 2 for schema and Step 1 ingestion of order.
- Step 4 depends on Step 1/2 for existing tables.
- Step 5 creates ordersummary before load (Step 6).
- Step 6 join depends on customer and order; loads to ordersummary.
- Step 7 defines ordersummary schema; used by SCD2 logic (Step 8).
- Step 8 depends on ordersummary and change_tracking_source customer.
- Step 9 creates customeraggregatespend before aggregation load (Step 10).
- Step 10 depends on ordersummary.
- Step 11 implements Steps 1–10 using Delta Live Tables.
- Name mismatches between lineage pairs: Not identified based on provided names.

13. Risks and Mitigations

- Risk: Ambiguity in deduplication keys. Mitigation: Define keys in implementation design. Information not present in source JSON.
- Risk: Null removal scope unclear. Mitigation: Specify columns/thresholds in technical design. Information not present in source JSON.
- Risk: SCD2 additional columns presence. Mitigation: Ensure RecordStatus, StartDate, EndDate are present in ordersummary physical schema.

14. Implementation Plan and Milestones

- Milestones:
- M1: Define Delta Live Tables pipeline with steps 1–10.
- M2: Implement schemas and ingestion.
- M3: Implement transformations and data quality.
- M4: Implement join and SCD2 logic.
- M5: Implement aggregation and target loads.
- M6: Validate outputs.
- Dates/owners: Information not present in source JSON.

15. Glossary

- SCD Type 2: Slowly Changing Dimension methodology that preserves historical records by closing out prior versions and creating new active versions.
- Delta Live Tables: Technology referenced for pipeline orchestration.
- Business Keys: Identifiers used to track SCD2 versions (CustId, OrderId).
- Not present in source JSON for any additional terms.

Appendix A: Repository and Technology Metadata

| Field | Value |

|---|---|

| Title | Data Processing Steps for Customer and Order Pipelines (Delta Live Tables) |

| Technology | Delta Live Tables |

| Catalog | gen_ai_poc_databrickscoe |

| Schema | sdlc_wizard |

Appendix B: Source Definitions and Inventory

| Source Name | Path | Format | Notes |

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| customer | /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata | csv | Not present in source JSON for additional details |

| order | /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata | csv | Not present in source JSON for additional details |

Appendix C: Target Definitions and Inventory

| Target Table | Catalog | Schema | Format | If Not Exists |

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| customer | Not present in source JSON | Not present in source JSON | delta | Not present in source JSON |

| order | Not present in source JSON | Not present in source JSON | delta | Not present in source JSON |

| ordersummary | gen_ai_poc_databrickscoe | sdlc_wizard | Not present in source JSON | true |

| customeraggregatespend | gen_ai_poc_databrickscoe | sdlc_wizard | Not present in source JSON | true |

Appendix D: Table Schemas

| Table | Column | Data Type | Notes |

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| customer | CustId | Not present in source JSON | From source JSON |

| customer | Name | Not present in source JSON | From source JSON |

| customer | EmailId | Not present in source JSON | From source JSON |

| customer | Region | Not present in source JSON | From source JSON |

| order | OrderId | Not present in source JSON | From source JSON |

| order | ItemName | Not present in source JSON | From source JSON |

| order | PricePerUnit | Not present in source JSON | From source JSON |

| order | Qty | Not present in source JSON | From source JSON |

| order | Date | Not present in source JSON | From source JSON |

| order | CustId | Not present in source JSON | From source JSON |

| ordersummary | CustId | Not present in source JSON | From source JSON |

ordersummary	Name	Not present in source JSON	From source JSON
ordersummary	EmailId	Not present in source JSON	From source JSON
ordersummary	Region	Not present in source JSON	From source JSON
ordersummary	OrderId	Not present in source JSON	From source JSON
ordersummary	ItemName	Not present in source JSON	From source JSON
ordersummary	PricePerUnit	Not present in source JSON	From source JSON
ordersummary	Qty	Not present in source JSON	From source JSON
ordersummary	Date	Not present in source JSON	From source JSON
ordersummary	RecordStatus	Not present in source JSON	Referenced in SCD2 rules
ordersummary	StartDate	Not present in source JSON	Referenced in SCD2 rules
ordersummary	EndDate	Not present in source JSON	Referenced in SCD2 rules
customeraggregatespend	Name	Not present in source JSON	From source JSON
customeraggregatespend	TotalAmount	Not present in source JSON	From source JSON
customeraggregatespend	Date	Not present in source JSON	From source JSON

Appendix E: Transformations

Table	Operation	Column	Expression	Notes
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order	add_columns	TotalAmount	PricePerUnit * Qty	From source JSON

Appendix F: Data Quality and Cleansing Rules

Rule Type	Tables	Details
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Remove Values	customer, order	Values to remove: "Null", null
Deduplication	customer, order	deduplicate = true

Appendix G: Joins and Relationships

Left Table	Right Table	Join Keys	Join Type	Target
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customer	order	CustId	inner	ordersummary

Appendix H: SCD Type 2 Configuration

Table	Business Keys	Change Tracking Source	Status Column	Status Active	Status Inactive	Start Column	End Column	Rules
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ordersummary	CustId, OrderId	customer	RecordStatus	Active	Inactive	StartDate	EndDate	When customer attributes change, mark old records as Inactive and new records as Active; Update StartDate and EndDate accordingly; Maintain full history in the table
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Appendix I: Aggregations and Measures

Source Table	Group By	Measure Name	Measure Expression	Target Table	Target Columns
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ordersummary	Name, Date	TotalAmount	sum(TotalAmount)	customeraggregatespend	Name, TotalAmount, Date
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Appendix J: Pipeline Flow, Steps, and Load Order

Step	Description	Depends On	Output/Target
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1	Read source CSV data from volumes and load into Delta tables	None	Delta tables for customer and order
2	Define schemas	1	Enforced schemas for customer and order
3	Add TotalAmount column to order	1,2	order with TotalAmount
4	Remove Null values and duplicates from both tables	1-3	Cleansed customer and order
5	Create the ordersummary table if it does not exist	1-4	ordersummary (empty or existing)
6	Join customer and order on CustId and load into SCD Type 2 table ordersummary	1-5	ordersummary populated
7	Define ordersummary schema	5-6	Enforced ordersummary schema
8	Implement SCD Type 2 logic for ordersummary	6-7	Historical tracking in ordersummary
9	Create the customeraggregatespend table if it does not exist	6-8	customeraggregatespend (empty or existing)
10	Aggregate TotalAmount grouped by Name and Date into customeraggregatespend	6-9	customeraggregatespend populated
11	Implement steps 1-10 using Delta Live Tables	1-10	Orchestrated pipeline

Appendix K: Parameters and Variables

Name	Value	Scope	Notes
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target_format	delta	Step 1	From source JSON
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scd_type	2	Step 6	From source JSON
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if_not_exists (ordersummary)	true	Step 5	From source JSON
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if_not_exists (customeraggregatespend)	true	Step 9	From source JSON
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Appendix L: Configuration Flags

Config	Value	Applies To
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deduplicate	true	Data quality for customer and order
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remove_values	"Null", null	Data quality for customer and order
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Appendix M: Environment and Deployment

Environment	Details
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Not present in source JSON	Not present in source JSON
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Appendix N: Error Handling and Logging

Aspect	Details
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Not present in source JSON	Not present in source JSON
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Appendix O: Security, Access Control, and Governance

Aspect	Details
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Not present in source JSON	Not present in source JSON
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Appendix P: Data Lineage Matrix

Source	Operation	Intermediate/Target	Notes
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| customer (CSV) | Ingest to Delta | customer (Delta) | Step 1 |

| order (CSV) | Ingest to Delta | order (Delta) | Step 1 |

| order | Add column | order.TotalAmount | Step 3 |

| customer, order | Inner join on CustId | ordersummary | Step 6 |

| ordersummary | SCD2 versioning | ordersummary (history) | Step 8 |

| ordersummary | Aggregate sum(TotalAmount) by Name, Date | customeraggregatespend | Step 10 |

Appendix Q: Naming Conventions and Mismatch Log

| Entity A | Entity B | Mismatch Type | Notes |

|---|---|---|---|

| Not present in source JSON | Not present in source JSON | Not present in source JSON | Not present in source JSON |

Appendix R: Scheduling and Orchestration

| Orchestrator | Schedule | Notes |

|---|---|---|

| Delta Live Tables | Not present in source JSON | Implements steps 1–10 |

Appendix S: Testing and Validation Scenarios

| Test ID | Objective | Input | Expected Output |

|---|---|---|---|

| T1 | Validate TotalAmount calculation | order with PricePerUnit and Qty | TotalAmount = PricePerUnit * Qty |

| T2 | Validate null removal | Records with "Null"/null | Records removed per rule |

| T3 | Validate deduplication | Duplicate customer/order rows | Duplicates removed |

| T4 | Validate join correctness | Matching CustId | Correct inner join result in ordersummary |

| T5 | Validate SCD2 behavior | Change in customer attributes | Prior record Inactive, new Active, dates updated |

| T6 | Validate aggregation | ordersummary with TotalAmount | Aggregated sums by Name, Date in customeraggregatespend |

Appendix T: Open Questions

ID	Question	Status
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Q1	What are data types for each column?	Not present in source JSON
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Q2	Which columns define deduplication uniqueness?	Not present in source JSON
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Q3	Environments (dev/test/prod) and deployment details?	Not present in source JSON
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Q4	Security and access controls for catalog/schema?	Not present in source JSON
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Q5	Scheduling cadence and SLAs?	Not present in source JSON
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Appendix U: Constraints and Limits

Constraint	Detail
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Source format	csv
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Target format	delta
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Technology	Delta Live Tables
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Appendix V: Expressions and Calculations

Expression Name	Table	Expression	Description
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TotalAmount Calculation	order	PricePerUnit * Qty	Computes order line extended amount
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Aggregation Measure	ordersummary	sum(TotalAmount)	Aggregates spend per customer per date
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Appendix W: Validation Columns and Status Flags

Table	Column	Purpose	Allowed Values
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ordersummary	RecordStatus	SCD2 active/inactive flag	Active, Inactive
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ordersummary	StartDate	SCD2 validity start	Not present in source JSON
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ordersummary	EndDate	SCD2 validity end	Not present in source JSON
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Appendix X: Performance and Scaling

Topic	Detail
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| Not present in source JSON | Not present in source JSON |

Appendix Y: Compliance and Regulatory

| Topic | Detail |

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| Not present in source JSON | Not present in source JSON |

Appendix Z: Change Log

| Version | Date | Description | Author |

|---|---|---|---|

| 1.0 | Not present in source JSON | Initial BRD derived from provided JSON | Not present in source JSON |